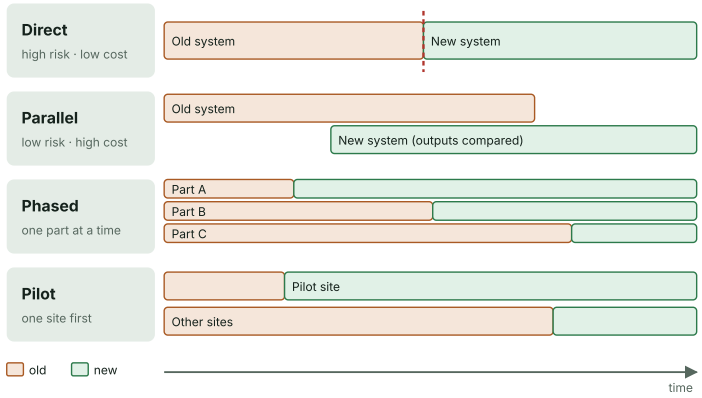
Implementation: changeover, data conversion and training

AP08 FACT FILE

Specification: students should be able to

- Describe the purpose of system implementation and the tasks it involves
- Describe and evaluate parallel, direct, pilot and phased changeover
- Recommend and justify a changeover method for a given scenario
- Explain what is meant by data conversion and describe what it involves
- Describe methods of training users and evaluate their suitability

Key words: implementation, changeover, parallel changeover, direct changeover, pilot changeover, phased changeover, big bang, data conversion, data migration, verification, installation, training, cascade training, e-learning, risk



Timelines comparing direct, parallel, phased and pilot changeover, showing when the old and new systems are in use

Overview: implementing a new system

Implementation is the stage at which a newly developed system is put into operation and replaces whatever was used before, whether an older computer system or a paper-based one. It has to be planned while the system is still being developed, because a lot has to be ready before the new system can take over completely: hardware and networks must be installed, software set up, staff trained and documentation provided. Existing data usually has to be converted into a form the new system can use. The organisation must then choose how to move from the old system to the new one. There are four changeover methods: parallel, where old and new run side by side; direct, where the old system stops and the new one starts on a set date; pilot, where the whole system is tried in one section first; and phased, where parts of the system are introduced one at a time. Each balances risk against cost and time differently, so the right choice depends on the organisation. After implementation, the system will need ongoing maintenance throughout its life.

Parallel and direct changeover

In parallel changeover the old and new systems operate together for a period. Every transaction is processed by both, and the two sets of results are checked against each other to confirm the new system gives correct output. When the new system has proved reliable, the old one is stopped. This is the safest method, because if the new system turns out to have flaws the old system is still there as a backup, and staff can become familiar with the new system gradually. However, it is expensive and time-consuming to keep two systems running, checking one against the other takes extra effort, and two separate copies of the data have to be kept, which may fall out of step. Direct changeover, sometimes called the big bang approach, switches off the old system on a particular date and introduces the new one immediately. There is no duplication of resources and everyone starts using the new system together. It is very risky, though: should the new system break down, going back to the old one may be difficult or impossible, and there may be a period when no system is available.

Pilot and phased changeover

Pilot changeover introduces the complete new system in one department, branch or section of the organisation first. The rest of the organisation keeps using the old system until the pilot has shown that the new one works, after which it is rolled out everywhere else. It suits organisations with several similar sections, such as a building society with many branches or a retailer with many shops. Only one section is affected if problems arise, and the staff in the pilot section gain experience that lets them help train others. Its drawbacks are that two systems run at the same time in different parts of the organisation, which needs extra resources, and it may be difficult to share data between sections using different systems. Phased changeover introduces the system one component at a time, for example payroll first, then invoicing, checking each part works before the next is switched on. Staff face change in stages, which can be less stressful, and a problem should affect only one part. However, the full changeover can take a long time, and requests to alter parts already introduced can delay the rest.

Choosing a changeover method

Because failure during changeover can be costly, organisations look for the lowest-risk method they can afford that is actually feasible. Where an organisation has several similar sites, pilot changeover is often a good balance, since it limits risk to one site without the full cost of parallel running. Where the consequences of failure would be severe, such as possible injury or major financial loss, parallel running may be justified despite its higher cost, because the old system remains as a safety net. Phased changeover suits a system made of distinct parts that can be introduced independently. Direct changeover carries the most risk and is used only when there is no realistic alternative, for example when the old and new systems cannot run at the same time, or for a small and non-critical system. A famous example of an unavoidable direct change was Sweden switching from driving on the left to driving on the right on a single morning in 1967: running both at once was impossible, so the change was planned in great detail months in advance. Direct changeover should be timed for a quiet period, with staff fully trained beforehand.

Data conversion

Data conversion is the task of transferring existing records into the structure and format the new system expects, so that it can be used fully from the start. When a computer system replaces a paper-based one, records such as forms, reports, orders and invoices must be turned into electronic data. This usually means creating the new database structure, employing staff to type in or scan the paper records, and then verifying and validating the data that has been entered. Verification might involve proof-reading entries against the original documents or keying data in twice, while validation rules reject impossible values. The volume of work can be large, particularly if scanned images of documents are to be stored. Where the new system replaces an older computer system, transferring the data, sometimes called data migration, is normally quicker, but fields may need reformatting, for example changing date formats or splitting a full name into first name and surname, and specialised or bespoke conversion software may be required. Poorly converted data causes errors and undermines users' confidence in the new system.

Training users

Users need training so that they can use the new system confidently and are more likely to accept it rather than resist it. Instructor-led classroom training allows questions and hands-on practice and gives everyone the same message, but it is expensive and takes staff away from their work. One-to-one or on-the-job training can be tailored to each person and uses real tasks, but it is slow and costly for large numbers. E-learning, video tutorials and online help let staff learn at their own pace and revisit material, and are cheap per person once produced, but offer no immediate help with problems and rely on staff being motivated. Printed manuals are cheap and useful for later reference but are often left unread. Cascade training, in which a few staff are trained and then train their colleagues, suits large organisations spread across many sites, although quality may fall at each level. Training must fit the changeover method: direct changeover needs everyone trained before the switch, whereas pilot staff can become the trainers for later sites.

Key terms from the mark schemes

Implementation
The stage at which a new system is put into operation, including installation, data conversion, training, documentation and changeover from the old system.

Changeover
The process of moving from an old system to a new system, using a planned method: parallel, direct, pilot or phased.

Parallel changeover
A changeover method in which the old and new systems run at the same time on the same transactions so that outputs can be compared until the new system is proven.
Credited: parallel running

Direct changeover
A changeover method in which the old system stops and the new one takes over at a single moment, with no overlap.
Credited: big bang / direct implementation

Pilot changeover
A changeover method in which the whole new system is introduced in one branch, department or site before being extended to every other part of the organisation.
Credited: pilot implementation

Phased changeover
A changeover method in which the new system is introduced one component or function at a time.
Credited: phased implementation

Data conversion
Transferring existing records into the structure and format the new system expects, including keying in or scanning paper records and verifying and validating the result.
Credited: data migration

Cascade training
A training method in which a small number of staff are trained and then train their colleagues.
Credited: train the trainer

Exam focus

Describe questions on changeover methods usually give a mark for what the method is and further marks for a benefit and a drawback, so learn at least one of each for all four methods. Evaluate or justify questions need you to apply advantages and disadvantages to the scenario and reach a reasoned recommendation; a general list with no link to the business earns limited credit. Phrase your answer around the organisation, for example explaining that parallel running is justified for payroll because an error would mean staff are paid wrongly. Classification questions describe a situation, often outside computing, and ask which method it shows. Look for the key feature: one location first means pilot, one part at a time means phased, both running together means parallel, and an overnight switch with no overlap means direct. The most common mistake is to confuse pilot and phased, so remember that pilot is the whole system in one place while phased is part of the system everywhere. For data conversion, refer to the actual records in the scenario, such as a diary or receipt book, and explain how they become tables in the new system.

? Check yourself

1. Describe the purpose of system implementation. [2]
2. Describe parallel changeover, giving one benefit and one drawback. [3]
3. Describe direct changeover, giving one benefit and one drawback. [3]

Answers: 1. To put the new system into operation/replace the old system (1) / ensuring resources such as trained staff, hardware and software are in place and data is converted so the system can become fully operational with minimum disruption (1) 2. Both systems operate at the same time, each transaction processed by both and outputs compared, until the new system is proven (1) / benefit: very safe, the old system is a backup if the new one fails / staff can get used to the new system (1) / drawback: expensive/time-consuming to run two systems / time needed to compare results / two sets of data may not be synchronised (1) 3. On a chosen date the old system stops and the new one takes over straight away, with no overlap (1) / benefit: no work is done twice / all staff switch at once / cheapest and quickest (1) / drawback: high risk, should the new system break down, going back may be hard/impossible / there may be a period with no system available (1)